

Estimating respiratory frequency by filtering Kinect v2 skeletal data

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Introduction: Monitoring a patient's breathing cycle is very important to identify early signs of serious complications manifested by decreased oxygen saturation, respiratory acidosis, and cardiac disturbances (e.g. cardiac arrest). This work presents a signal processing technique to measure respiratory frequency by relying only the skeletal tracking data of an inexpensive depth camera. Our approach aims to be useful for reading the breath frequency automatically in a non-invasive and markerless manner. This study applies a pass-band filter that magnifies the micro-movements (1) of skeleton's joints to compute the respiratory frequency.

Materials and Methods: The Microsoft Kinect v2 was used as a depth camera which computes a skeletal representation of the human body, which consists of 25 notable points (joints and body extremities). Here, we only consider the 5 torso points to analyze the breathing movement. To evaluate the system, we collected data from 6 subjects along 4 acquisitions of during 60 seconds each: seated with a normal breath; standing with a normal breath; seated with a heavy breath; and standing with a heavy breath. To assess our approach, we considered as ground truth data acquired from the BioSignalsPlux Researcher device. Informed consent was received from 6 volunteers (2 female, 4 male) with no history of lung disease or pain.

Results: The results show that it is possible to extract reliable kinematic data from skeletal data to calculate the respiratory rate. Figure 1 shows the differences between the estimated frequency and the data acquired from the BioSignalsPlux device.

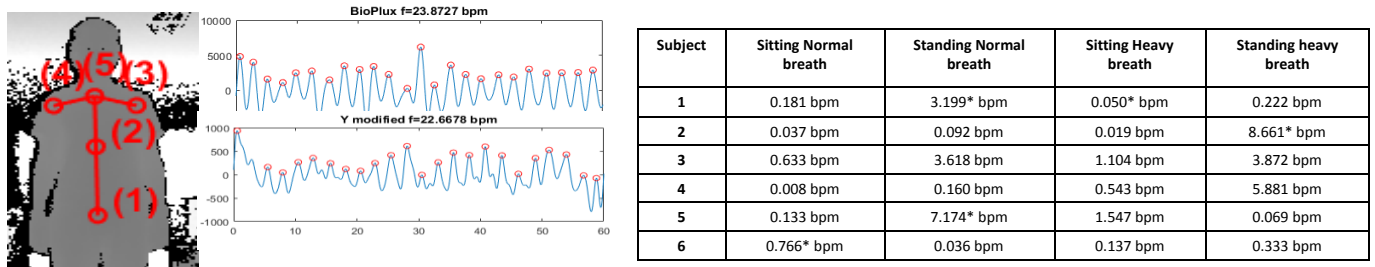


Figure 1 (Left) Skeletal data with 5 joints superimposed to the depth image. (Center) Plots of a normal breath cycle acquired with BioSignalsPlux (top) and with the Kinect v2 (bottom). (Right) Frequency breath difference between BioSignalsPlux and the closest result from the most representative point coordinate within the skeletal joints (*Unsatisfactory BioSignalsPlux results; bpm – breaths per minute)

Discussion and Conclusions: The preliminary results are very promising, although relatively high discrepancies still persist and should not be ignored. Further acquisitions with many more subjects (e.g. dozens of participants) and in different settings (at rest, laid down, seated, standing, after moderate and intense exercises) are required to attain statistical significance. Still, the results reveal that it is possible to effectively acquire the respiratory rate via skeletal data from Kinect v2. In addition, more skeletal points need to be processed to properly map which points and point coordinate contains the most representative physiological signal.

References:

[1] H.-Y. Wu, M. Rubinstein, E. Shih, J. Guttag, F. Durand, and W. Freeman, Eulerian video magnification for revealing subtle changes in the world. *ACM Trans. Graph.*, vol. 31, no. 4, pp. 1–8, 2012.

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